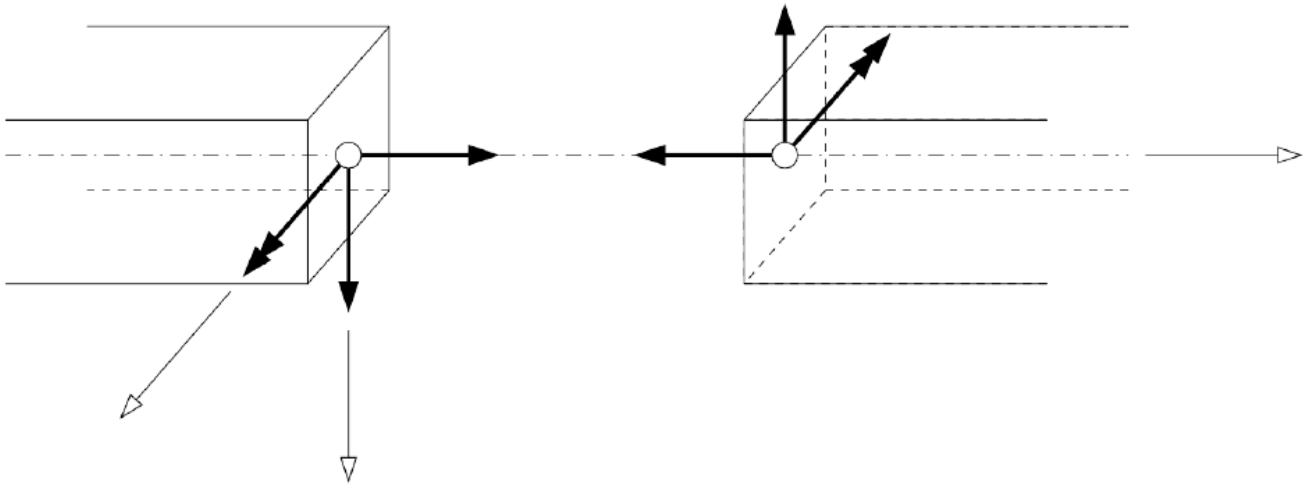
Engineering Mechanics I

Lecture



Autonomous Aerial Systems

7.7. Practical procedure for structural analysis

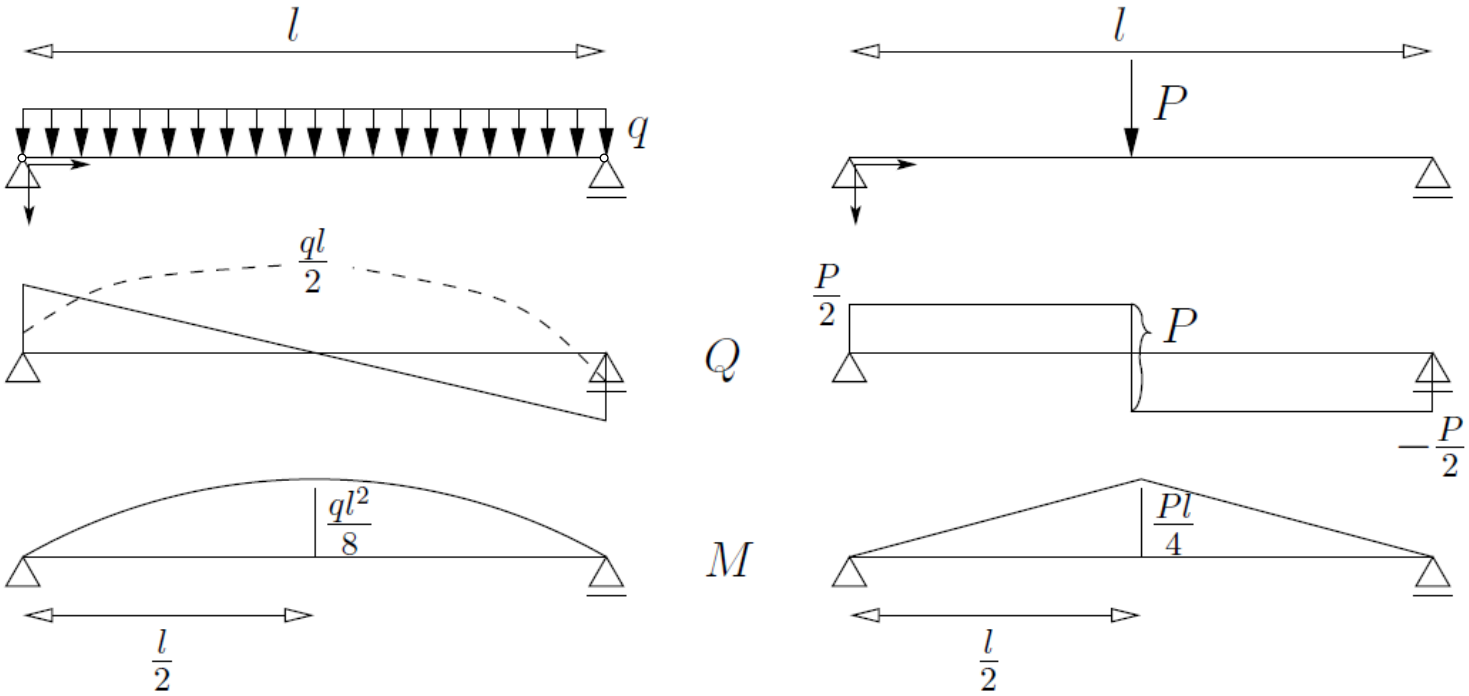
- **Integration method** is error-prone, error propagation!

Alternative:

- Calculate internal forces via **equilibrium** at designated points.
- **Progression** results from knowledge of the differential relations.
- The course is "hooked" into the graphical representation.
- Superposition principle allows superposition.

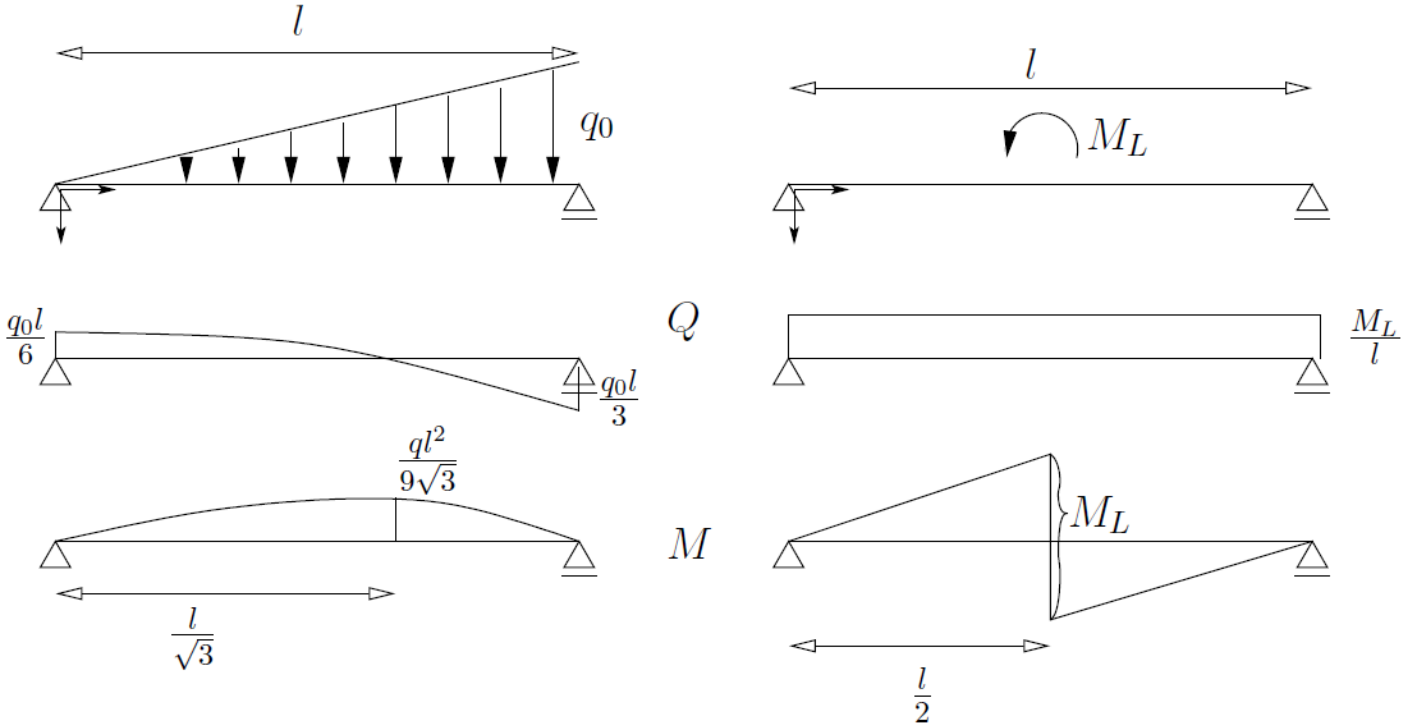
7.7. Practical procedure for structural analysis

Internal forces for most important cases



7.7. Practical procedure for structural analysis

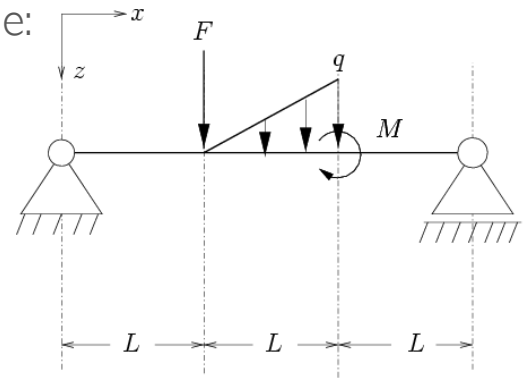
Internal forces for most important cases

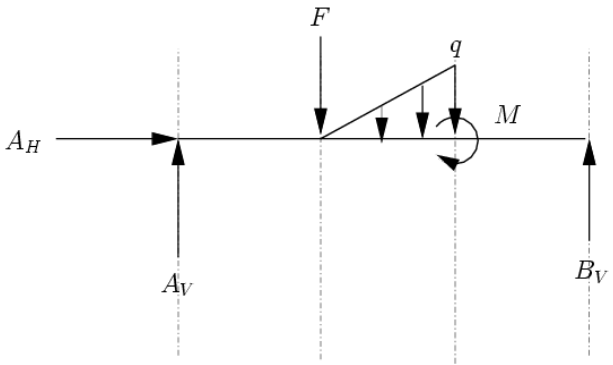


7.7. Practical procedure for structural analysis

Illustration by example

- Example:





- Support Reactions: $M := 2\,FL$ and $q_0 L := 3F$

$$B_V = \frac{F}{3} + \frac{5}{18} q_0 L + \frac{M}{3L}, \qquad A_V = \frac{2}{3} F + \frac{2}{9} q_0 L - \frac{M}{3L}$$

with $M := 2FL$ and $q_0 := \frac{3F}{L}$ $\quad \rightarrowtail \quad$ $B_V = \frac{11}{6} F, \quad A_V = \frac{2}{3} F$

7.7. Practical procedure for structural analysis

- Section 1:

Box 7.29

- Section 2:

Box 7.30

7.7. Practical procedure for structural analysis

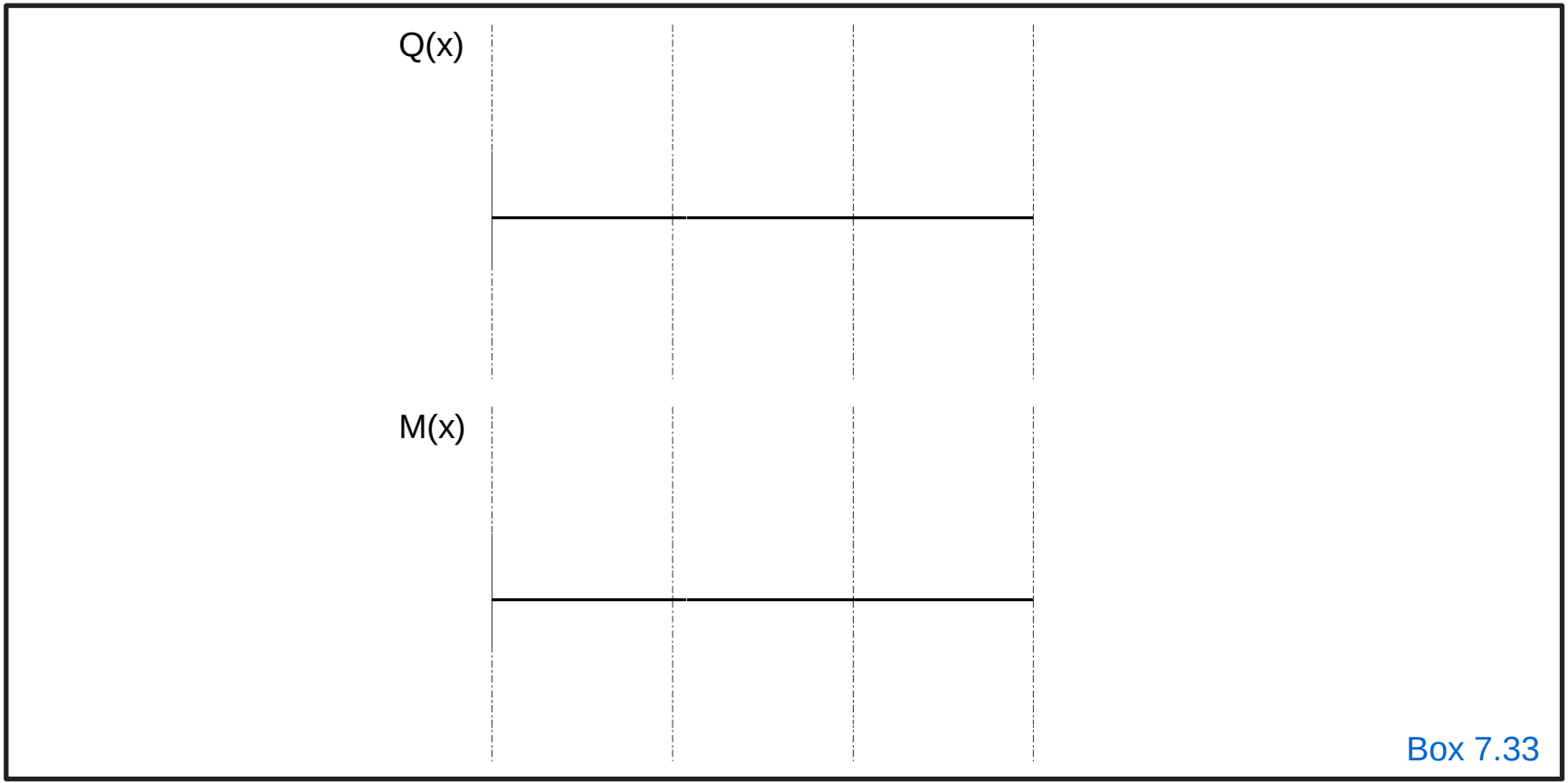
- Section 2 (coming from right side):

Box 7.31

- Section 3 (coming from right side):

Box 7.32

7.7. Practical procedure for structural analysis



Box 7.33

Autonomous Aerial Systems

8. Frames and Arches

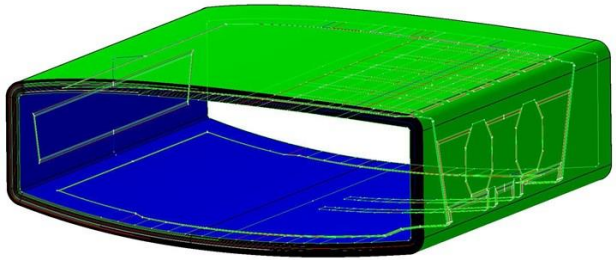






Autonomous Aerial Systems

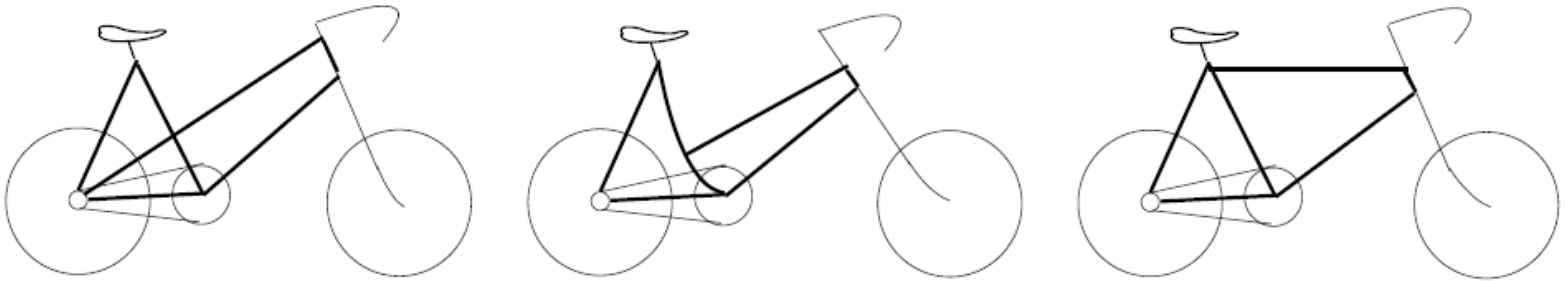
8. Frames and Arches







8. Frames and Arches



Frames:

- Beam structures with kinks and/or bifurcation.
- Statically determinate frames can be subdivided into beam sections and calculated according to the previous section.
- Transition conditions must be considered at kinks and bifurcations

8.1. Frames

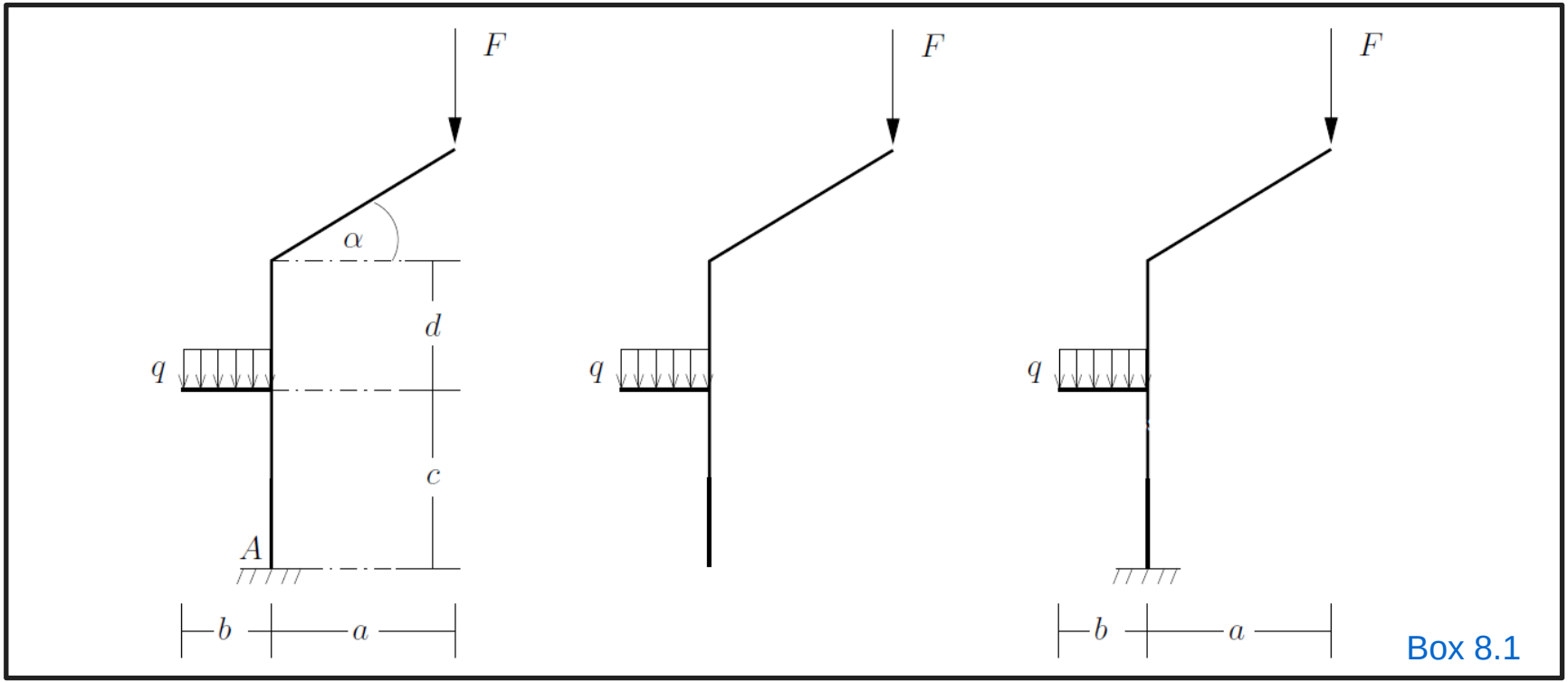
Rigid frame

Procedure:

- Support reactions
- Section subdivision
- Coordinate system for each section, which remains binding for internal forces
- Cut out kinks and bifurcations
- Establish equilibrium conditions
- Work from the "outside" to the "inside" of the structure
- Suitable sections avoid errors

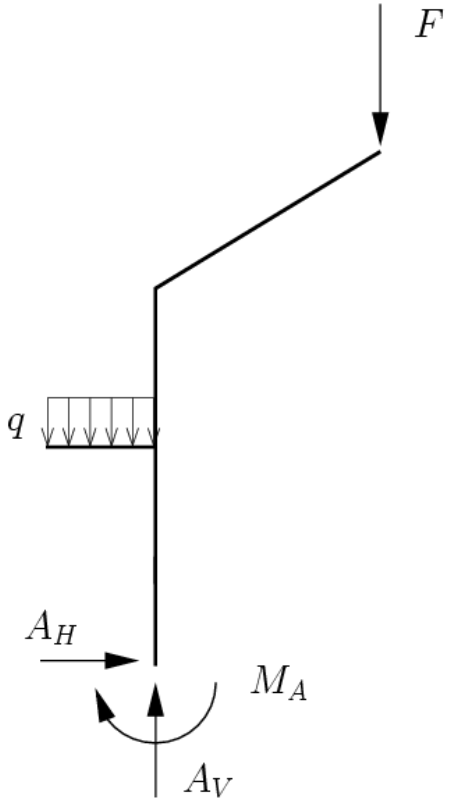
8.1. Frames

- Frame example with kink and bifurcation



Box 8.1

8.1. Frames



Support Reactions

Box 8.2

8.1. Frames

Course known from differential relations. Calculation only at distinguished places!

- Section 4, Cut at Support:

Resulting Forces:

- Normal Force:
- Shear Force:
- Moment:

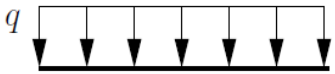
Box 8.3

8.1. Frames

- Section 3 with **line load**:

Resulting forces:

- Normal Force: $N_3(b) = 0$
- Shear Force: $Q_3(b) = -qb$
- Moment:



Box 8.4